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United States Patent	12401210
Kind Code	B2
Date of Patent	August 26, 2025
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Rectification by battery protection system

Abstract

In some examples, a controller circuit comprises: a voltage subtractor circuit having a subtractor output and first and second subtractor inputs, the first subtractor input adapted to be coupled to a first current terminal of a transistor, and the second subtractor input adapted to be coupled to a second current terminal of the transistor; an internal voltage generator circuit having a generator input and a generator output, the generator input adapted to be coupled to the first current terminal; and a gate control circuit having a gate control input and a gate control output, the gate control input coupled to the subtractor output, the gate control output adapted to be coupled to a gate of the transistor, the gate control circuit including a switch coupled between the gate control output and the generator output.

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Appl. No.:	18/215922
Filed:	June 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230336009 A1	Oct. 19, 2023

Related U.S. Application Data

continuation parent-doc US 17513746 20211028 US 11735933 child-doc US 18215922

Publication Classification

Int. Cl.: H02J7/00 (20060101)

U.S. Cl.:

CPC H02J7/0034 (20130101); H02J7/0031 (20130101); H02J7/0047 (20130101); H02J7/0063 (20130101); H02J7/00711 (20200101)

Field of Classification Search

CPC: H02J (7/00309); H02J (7/00711); H02J (7/007186); H02J (7/0047)

USPC: 361/84

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Background/Summary

(1) This application is a continuation of application Ser. No. 17/513,746 filed Oct. 28, 2021, which is incorporated herein by reference.

BACKGROUND

(1) An electric power system that transfers electric power from a battery to a load may include a protection system to protect the load from a reverse battery connection, where the load may receive a negative input voltage from the battery. The protection system can isolate the load from the negative input voltage to prevent the load from being damaged by the negative input voltage. Some examples of the protection system can also block a reverse current from flowing from the load to the battery, to allow the load side additional time to operate before turning off. While the battery may output a direct current (DC) signal, the battery outputs may also include alternating current (AC) signals superimposed on the DC signal. It is also desirable that the protection system can have a short response time, so that the protection system can properly and promptly respond to transient signals output by the battery to protect the load.

SUMMARY

(2) A controller circuit includes a voltage subtractor circuit, an internal voltage generator circuit, and a gate control circuit. The voltage subtractor circuit has a subtractor output and first and second subtractor inputs. The first subtractor input is adapted to be coupled to a first current terminal of a

transistor. The second subtractor input is adapted to be coupled to a second current terminal of the transistor. The internal voltage generator circuit has a generator input and a generator output. The generator input is adapted to be coupled to the first current terminal. The gate control circuit has a first gate control input, a second gate control input, and a gate control output. The first gate control input is coupled to the subtractor output. The second gate control input is coupled to the generator output. The gate control output is adapted to be coupled to a gate of the transistor. The gate control circuit also includes a switch coupled between the gate control output and the second gate control input.

(3) In a method, a first voltage is received via a first terminal of a controller circuit, the first terminal being coupled to a first current terminal of a transistor. A second voltage is received via a second terminal of the controller circuit, the second terminal being coupled to a second diffusion of the transistor. Based on the first voltage and the second voltage, a switch between a voltage reference and a third terminal of the controller circuit coupled to a gate of the transistor is closed. A voltage of the gate is set by the voltage reference to form a conduction channel between the first current terminal and the second current terminal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic diagram of an electric power transfer system in accordance with various examples.

(2) FIGS. 2A through 2F illustrate examples of internal components of a protection system and their operations.

(3) FIGS. 3A, 3B, and 3C illustrate examples of rectification operations performed by the protection system of FIG. 2A-FIG. 2F.

(4) FIGS. 4A through 4F illustrate examples of a protection system including internal components to perform a rectification operation in accordance with various examples.

(5) FIG. 5 illustrates examples of voltage graphs depicting a rectification operation by the protection system of FIGS. 4A through 4F.

(6) FIG. 6 is a flowchart of an example method for performing a rectification operation in accordance with various examples.

DETAILED DESCRIPTION

(7) As described above, an electric power system may include a protection system to protect the load from a reverse battery connection. In a case where the electric power system is part of a vehicle, reverse battery connection may occur during maintenance of the vehicle's battery or jump start of the vehicle. Without the protection system, the load may receive a negative voltage from the battery when the battery is reversely connected. The negative voltage can cause huge current to flow from various electronic components of the load, such as electrostatic discharge (ESD) circuits, voltage regulators, etc., which can cause severe damage to these components.

(8) The protection system can include a controller circuit and a transistor. The transistor can include a body diode, of which the anode can be coupled to the battery and the cathode can be coupled to the load. In a reverse battery connection, the battery may output a negative voltage, and the controller circuit can disable the transistor and rely on the reversed-bias body diode to isolate the load from the negative voltage, and to prevent a reverse current from flowing from the load back to the battery. If the battery is connected in the correct polarities, the controller circuit can enable the transistor to transmit a positive voltage and a forward current from the battery to the load. In a case where the anode receives a voltage including both DC and AC components, the protection system can perform a rectification operation to transmit positive AC components (e.g., AC components that adds to the DC component) and not to transmit negative AC components (e.g., AC components

that subtracts from the DC component) to the load. For reasons to be described below, the controller circuit may incur substantial delay to enable the transistor, which makes it difficult to perform rectification operations for high frequency AC components. Example techniques described herein speed up the rectification response of the controller circuit, which allows the protection system to perform rectification operations for AC components at a high frequency (e.g., 200 kilohertz (kHz) or above).

(9) FIG. 1 illustrates an example of a system **100**. System **100** may include a battery **102**, an electric power system **104**, and a load **106**. Load **106** may include an internal power supply **108**, which can include a linear regulator (e.g., a low dropout regulator) and/or a switch mode regulator (e.g., a buck converter, a boost converter, a buck-boost converter, etc.) to provide a supply voltage, and a holdup capacitor to supply a current. Load **106** may further include subsystems **110** that draw power from internal power supply **108**. Both internal power supply **108** and subsystems **110** can include various electronic components.

(10) Electric power system **104** is configured to transfer electric power from battery **102** to load **106**. Electric power system **104** may receive a voltage $V_{sub.in}$ and a current $I_{sub.in}$, and provide a voltage $V_{sub.out}$ and a current $I_{sub.out}$ to load **106**. Internal power supply **108** can receive voltage $V_{sub.out}$ and current $I_{sub.out}$ from electric power system **104** and provide a voltage $V_{sub.out_internal}$ and a current $I_{sub.out_internal}$ to subsystems **110**. Voltage $V_{sub.out}$ and current $I_{sub.out}$ provided by electric power system **104** can be based on, respectively, voltage $V_{sub.in}$ and current $I_{sub.in}$ provided by battery **102**. Also, voltage $V_{sub.out_internal}$ and current $I_{sub.out_internal}$ can be based on a configuration of internal power supply **108** and subsystems **110**. For example, $V_{sub.out_internal}$ can be a fraction of $V_{sub.out}$ to provide a reduced supply voltage required by subsystems **110**, and $I_{sub.out_internal}$ can be reduced from $I_{sub.in}$ due to power consumption by electric power system **104** and internal power supply **108**. In FIG. 1, the positive terminal of battery **102** can be coupled to electric power system **104** to supply a positive voltage $V_{sub.in}$, while the negative terminal of battery **102** can be coupled to ground. With such configuration, voltages $V_{sub.out}$ and $V_{sub.out_internal}$ can be positive, and currents $I_{sub.in}$, $I_{sub.out}$, and $I_{sub.out_internal}$ can be part of a forward current that flows from battery **102** to load **106**.

(11) In some examples, electric power system **104** can include a reverse battery protection system **112** to protect load **106** from a reverse battery connection, where the positive terminal of battery **102** is coupled to ground and the negative terminal of battery **102** is coupled to electric power system **104**. As a result, battery **102** may transmit a negative voltage, such as $-V_{sub.in}$, to electric power system **104**. Without reverse battery protection system **112**, electric power system may transmit the negative voltage to load **106**. The negative voltage can cause a huge current to flow from various electronic components of load **106**, such as electrostatic discharge (ESD) circuits, voltage regulators of internal power supply **108**, etc., which can cause severe damage to these components. Moreover, a reverse current may also flow from load **106** back to battery **102**. The reverse current may discharge the holdup capacitor of internal power supply **108** and reduces the holdup capacitor's capability of supplying power to subsystems **110**. Reverse battery protection system **112** can isolate load **106** from the negative voltage $-V_{sub.in}$. In some examples, reverse battery protection system **112** can also block the reverse current from flowing from load **106** to battery **102**, to allow subsystems **110** additional time to operate before turning off.

(12) FIGS. 2A through 2F illustrate examples of internal components of reverse battery protection system **112** of system **100** and their operations. Referring to the left side of FIG. 2A, reverse battery protection system **112** can include a controller circuit **200** and a transistor **202**. Transistor **202** can be an n-channel FET (NFET) or a p-channel FET (PFET). Transistor **202** can have a gate **204**, a first current terminal **206**, and a second current terminal **208**. A body diode **210** can be formed at a p-n junction between first current terminal **206** and second current terminal **208**, with first current terminal **206** being an anode (denoted "A" in the figures) and second current terminal **208** being a

cathode (denoted “C” in the figures). In a case where transistor **202** is an NFET, first current terminal **206** can be a source whereas second current terminal **208** can be a drain. In a case where transistor **202** is a PFET, first current terminal **206** can be a drain whereas second current terminal **208** can be a source. In system **100**, first current terminal **206** can be coupled to battery **102** at a node **220**, and second current terminal **208** can be coupled to load **106** at a node **222**.

(13) In FIG. 2A, system **100** may include a capacitor **224** and a capacitor **226**. Capacitor **224** can model a combination of parasitic capacitances at node **220**, such as capacitances of wires and electrical connectors between battery **102** and transistor **202**, the junction capacitance at first current terminal **206**, etc. Moreover, capacitor **226** can model a combination of parasitic capacitances at node **222**, such as capacitances of wires and electrical connectors between load **106** and transistor **202**, the junction capacitance at second current terminal **208**, etc. Capacitor **226** can also include a physical hold up capacitor to provide a temporary power supply to load **106** when transistor **202** is disabled.

(14) Transistor **202** can be coupled to and controlled by controller circuit **200** to emulate an ideal diode having the same anode and cathode as body diode **210**. In some examples, controller circuit **200** can include a terminal **230** adapted to be coupled to first current terminal **206**. First current terminal **206** can be the anode of the ideal diode. Controller circuit **200** can also include a terminal **232** adapted to be coupled to gate **204** of transistor **202**, and a terminal **234** adapted to be coupled to second current terminal **208** of transistor **202**. Second current terminal **208** can be the cathode of the ideal diode. Terminals **230**, **232**, and **234** can include interconnects (e.g., chip-chip interconnects, traces on printed circuit board (PCB), etc.) that allow signals (e.g., current, voltage, etc.) to flow between controller circuit **200** and transistor **202**. Controller circuit **200** can monitor the anode voltage $V_{sub.A}$ at first current terminal **206** and the cathode voltage $V_{sub.C}$ at second current terminal **208**, and adjust the voltage of gate **204** of transistor **202** via terminal **232** responsive to changes of the anode-cathode voltage $V_{sub.AC}$ to emulate an ideal diode coupled between battery **102** and load **106**.

(15) The right side of FIG. 2A illustrates an example transfer function graph **212** of an ideal diode to be emulated by transistor **202**. Transfer function graph **212** illustrates a relationship between the amount of a forward current $I_{sub.F}$ conducted by the diode, from anode to cathode, with respect a difference voltage between the anode and cathode $V_{sub.AC}$. If $V_{sub.AC}$ is below a forward voltage $V_{sub.F}$, the diode can be reverse-biased, and no forward current (or a minimum amount of forward current) flows through the diode. If $V_{sub.AC}$ is above the forward voltage $V_{sub.F}$, the diode is forward bias and can conduct a forward current $I_{sub.F}$. When the diode is forward-biased, the anode-cathode voltage $V_{sub.AC}$ can remain constant at $V_{sub.F}$ independent of the amount of forward current $I_{sub.F}$ being conducted, so that the cathode voltage $V_{sub.C}$ can be equal to the anode voltage $V_{sub.A}$ minus the forward voltage $V_{sub.F}$.

(16) To emulate the ideal diode, in a case where $V_{sub.AC}$ is above a forward voltage threshold representing the forward voltage of the ideal diode, controller circuit **200** can increase the gate-source voltage ($V_{sub.GS}$) of transistor **202** (if transistor **202** is an NFET), or the source-gate voltage ($V_{sub.SG}$) of transistor **202** (if transistor **202** is a PFET), to be above a threshold voltage $V_{sub.th}$ of the transistor. Raising $V_{sub.GS}$ ($V_{sub.SG}$) to above $V_{sub.th}$ can turn on/enable transistor **202** by forming a conduction channel between first current terminal **206** and second current terminal **208** under gate **204**. The conduction channel can transmit a positive voltage and a forward/positive current from battery **102** to load **106**. However, in a case where $V_{sub.AC}$ is below the forward voltage threshold, controller circuit **200** can reduce the gate-source voltage $V_{sub.GS}$ (if transistor **202** is NFET) or source-gate voltage $V_{sub.SG}$ (if transistor **202** is PFET) to be below the threshold voltage $V_{sub.th}$. Dropping $V_{sub.GS}$ (or $V_{sub.SG}$) below $V_{sub.th}$ can turn off/disable transistor **202** by removing (or at least reducing) the conduction channel. Body diode **210** is reverse-biased due to $V_{sub.AC}$ being below the forward voltage threshold, and the reverse-biased body diode can block a negative voltage and a reverse/negative current from reaching load

106 from battery **102**.

(17) Although transfer function graph **212** shows that an ideal diode has a single forward voltage $V_{sub.F}$, in some examples controller circuit **200** can enable a conduction channel of transistor **202** (between first current terminal **206** and second current terminal **208**) in response to $V_{sub.AC}$ exceeding multiple thresholds, which can indicate that the battery is connected with the correct polarity. Controller circuit **200** can also disable/remove the conduction channel of transistor **202** to block a reverse current/negative voltage in response to $V_{sub.AC}$ being below a reverse bias threshold, which can indicate a reverse battery connection. Such arrangements can improve the robustness of system **100** in light of transient noises.

(18) FIG. 2B illustrates a flowchart of an example method **240** performed by controller circuit **200** in controlling transistor **202**. Method **240** can be performed after controller circuit **200** starts up and has not yet started enabled transistor **202**.

(19) In step **241**, controller circuit **200** can determine an anode-cathode voltage ($V_{sub.ac}$) across transistor **202**. Controller circuit **200** can monitor the anode voltage ($V_{sub.A}$) at terminal **230** and the cathode voltage ($V_{sub.C}$) at terminal **234**. Controller circuit **200** can include a subtraction circuit (e.g., implemented using a differential amplifier) to subtract $V_{sub.C}$ from $V_{sub.A}$ to obtain $V_{sub.AC}$.

(20) Controller circuit **200** can then proceed to compare $V_{sub.AC}$ with a forward conduction threshold voltage $V_{sub.F-on}$, in step **242**. If $V_{sub.AC}$ exceeds $V_{sub.F-on}$, controller circuit **200** can start a regulation loop to raise the gate-source voltage $V_{sub.GS}$ (or $V_{sub.SG}$ if transistor **202** is PFET) to enable a conduction channel of transistor **202**, and to regulate $V_{sub.AC}$ at a target forward voltage $V_{sub.F-reg}$, in step **243**. $V_{sub.F-reg}$ can represent $V_{sub.F}$ of an ideal diode in transfer function graph **212** of FIG. 2A, and transistor **202** can be controlled to emulate a forward-biased diode. In step **244**, $V_{sub.AC}$ reaches (and can be regulated) at $V_{sub.F-reg}$.

(21) The forward conduction threshold voltage $V_{sub.F-on}$ can be made higher than $V_{sub.F-reg}$. By having $V_{sub.AC}$ to be higher than $V_{sub.F-on}$ (and to be much higher than $V_{sub.F-reg}$) to start the forward conduction, the likelihood of mistaking a transient noise at node **220** as a positive voltage supplied by battery **102**, and falsely enabling transistor **202** as a result, can be reduced. The target forward voltage $V_{sub.F-reg}$ can be regulated at a lower voltage than $V_{sub.F-on}$ to reduce voltage drop and power loss across transistor **202** when emulating the forward-biased diode.

(22) Also, controller circuit **200** can compare $V_{sub.AC}$ with a reverse bias threshold voltage $V_{sub.R}$, in step **246**, to detect a reverse battery connection. The reverse bias threshold voltage $V_{sub.R}$ can be a negative voltage that can be received from the negative terminal of battery **102** when the polarity of battery **102** is reversed. Therefore, comparing $V_{sub.AC}$ against a negative voltage to detect a reverse battery connection can reduce the likelihood of false detection of reverse battery connection, such as caused by a transient voltage at node **220**. If $V_{sub.AC}$ is below $V_{sub.R}$, which can indicate a reverse battery connection, or if $V_{sub.AC}$ is above $V_{sub.R}$ but below $V_{sub.F-on}$, which can indicate a small transient voltage rather than a large positive voltage supplied by battery **102**, controller circuit **200** can maintain transistor **202** in a disabled state, in step **247**. In a case where transistor **202** is disabled and the conduction channel is removed, the reverse-biased body diode **210** can block a negative voltage/a reverse current.

(23) FIG. 2C illustrates examples of internal components of controller circuit **200**. Referring to FIG. 2B, controller circuit **200** can include a gate control circuit **250**. Gate control circuit **250** can include an input **251**, an input **252**, and an output **253**. Input **251** can be adapted to be coupled to first current terminal **206** of transistor **202**, which can be the anode of the diode to be emulated, via terminal **230**. Output **253** can be adapted to be coupled to gate **204** via terminal **232**. Controller circuit **200** further includes a voltage subtractor circuit **254**, which can include an op-amp subtractor or other suitable circuits, to receive an anode voltage $V_{sub.A}$ via terminal **230** and a cathode voltage $V_{sub.C}$ via terminal **234**, generate an anode-cathode voltage $V_{sub.AC}$ representing a difference between $V_{sub.A}$ and $V_{sub.C}$, and provide $V_{sub.AC}$ to input **252** of gate

control circuit **250**. Gate control circuit **250** can generate a gate voltage signal $V_{sub.G}$ in response to $V_{sub.AC}$, based on the techniques described in method **240** of FIG. 2B, and provide gate voltage signal $V_{sub.G}$ via output **253** and terminal **232** to enable transistor **202** to conduct a forward current from the anode to the cathode (and from battery **102** to load **106**), or to disable transistor **202** to block the flow of a reverse current from the cathode back to the anode (and from load **106** back to battery **102**).

(24) Also, controller circuit **200** can include a local voltage generator circuit **256** to generate local voltages. Local voltage generator circuit **256** can receive the anode voltage, which can be a positive voltage provided by battery **102**, via terminal **230** as an input ($V_{sub.in}$). Local voltage generator circuit **256** can provide a high supply voltage ($V_{sub.h}$) to a high power supply terminal (labelled “PWRH” in FIG. 2C) of gate control circuit **250**, and a low supply voltage (V_i) to a low power supply terminal (labelled “PWRL” in FIG. 2C) of gate control circuit **250**. The high supply voltage and the low supply voltage can be generated from the anode voltage $V_{sub.A}$ and supplied to gate control circuit **250** to reduce the drain-source voltages ($V_{sub.DS}$) of devices of gate control circuit **250** and the resulting voltage stress. Local voltage generator circuit **256** can include a charge pump to generate the high supply voltage $V_{sub.h}$ by adding an offset voltage to the anode voltage $V_{sub.A}$. Local voltage generator circuit **256** can also include a linear regulator, such as a floating-rail low drop out (LDO) regulator, to generate the low supply voltage $V_{sub.l}$ by subtracting an offset voltage from the anode voltage $V_{sub.A}$.

(25) FIG. 2D illustrates examples of internal components of gate control circuit **250**. Referring to FIG. 2D, gate control circuit **250** can include a reverse current blocking (RCB) circuit **260**, which can include a network of comparators including comparators **262a** and **262b**, an RCB logic circuit **264**, and a switch **266**. Switch **266** is coupled between input **251** (which can be coupled to first current terminal **206**/anode of transistor **202**) and output **253** (which can be coupled to gate **204**). Gate control circuit **250** can also include a forward conduction control circuit **270**, which can include an amplifier **272**, such as an operational transconductance amplifier (OTA), an op-amp, etc., and a switch **274**. Switch **274** can be coupled between the output of amplifier **272** and output **253**. In some examples, switch **274** can be part of a switchable output stage of amplifier **272**. RCB logic **264** can control switches **266** and **274** via a pair of complimentary control signals **280a** and **280b**. Accordingly, when switch **266** is closed, switch **274** can be opened, and vice versa. Gate control circuit **250** can include an inverter **282** to generate control signal **280b** from control signal **280a**.

(26) RCB circuit **260** and forward conduction control circuit **270**, through switches **266** and **274**, can set the gate-source voltage $V_{sub.GS}$ of transistor **202** in response to the anode-cathode voltage $V_{sub.AC}$, to enable the flow of a forward current from the anode to the cathode (and from battery **102** to load **106**), and to block the flow of a reverse current from the cathode back to the anode (and from load **106** back to battery **102**), based on techniques described in FIG. 2B.

(27) Specifically, referring to FIG. 2E, each of comparators **262a** and **262b** can receive $V_{sub.AC}$ from voltage subtractor circuit **254**. Comparator **262a** can compare $V_{sub.AC}$ against forward conduction threshold voltage $V_{sub.F-on}$ to generate a first decision, and comparator **262b** can compare $V_{sub.AC}$ against reverse bias threshold voltage $V_{sub.R}$ to generate a second decision. If $V_{sub.AC}$ is below $V_{sub.F-on}$ (which can indicate $V_{sub.AC}$ is raised by a small transient voltage), or if $V_{sub.AC}$ is below $V_{sub.R}$ (which can indicate a reverse battery connection), RCB logic **264** can provide a control signal **282a** to disable transistor **202**. Controller circuit **200** can generate control signal **280a** to close switch **266** to connect first current terminal **206** with gate **204**. By connecting first current terminal **206** with gate **204**, the gate voltage $V_{sub.G}$ can be set to be equal to the source voltage $V_{sub.S}$, and the gate-source voltage ($V_{sub.GS}$) for transistor **202** can be reduced to zero. With the $V_{sub.GS}$ voltage below a threshold voltage $V_{sub.th}$ for forming a channel below gate **204**, transistor **202** can be disabled, and the flow of current between first current terminal **206** and second current terminal **208** of transistor **202** can also be disabled.

Moreover, inverter **282** can generate control signal **280b** as a complimentary version of control signal **280a** to open switch **274**, and the output of amplifier **272** can be disconnected from output **253** (and terminal **232**) to avoid interfering with the setting of the gate-source voltage ($V_{sub.GS}$) for transistor **202** by RCB circuit **260**.

(28) In some examples, RCB logic circuit **264** can include a timing circuit, such as a timer. RCB logic circuit **264** can start the timer after disabling switch **266**. The timer can define an RCB timing window in which transistor **202** is to be continuously disabled regardless of whether $V_{sub.AC}$ is below or above the forward conduction threshold voltage $V_{sub.F-on}$, and switch **266** is to be continuously enabled. Within the RCB timing window, RCB logic circuit **264** can ignore decisions from comparators **262a** and **262b** to continue closing switch **266** to disable transistor **202**, and continue opening switch **274** to disconnect the output of amplifier **272** from gate **204**. Such arrangements can reduce the likelihood of controller circuit **200** falsely starting a forward conduction due to transient signals at the anode/cathode. The duration of the RCB timing window can be fixed (e.g., built into RCB logic circuit **264**) or can be programmable via a register coupled to RCB logic circuit **264** (not shown in the figures).

(29) FIG. 2F illustrates examples of operations of gate control circuit **250** to enable forward conduction. Referring to FIG. 2F, if the decisions of comparators **262a** and **262b** indicate that $V_{sub.AC}$ is higher than $V_{sub.F-on}$, which can indicate that the battery is connected in the correct polarities (e.g., positive terminal being coupled to the anode of transistor **202**), RCB logic **264** can generate control signal **280a** to open switch **266**, while control signal **280b**, being a complimentary version of control signal **280a**, can close switch **274** to connect the output of amplifier **272** with output **253** (and gate **204** via terminal **232**). Amplifier **272** is then allowed to adjust the gate voltage $V_{sub.G}$ (or decrease the gate voltage $V_{sub.G}$ if transistor **202** is PFET) via output **253** and terminal **232**. With the anode voltage $V_{sub.A}$ largely fixed by battery **102**, if the gate-source voltage $V_{sub.GS}$ (or $V_{sub.SG}$ if transistor **202** is PFET) becomes higher than the threshold voltage $V_{sub.th}$ of transistor **202**, a conduction channel can be created between first current terminal **206** and second current terminal **208** of transistor **202**. The conduction channel can then enable the flow of forward current $I_{sub.F}$ from first current terminal **206** to second current terminal **208** of transistor **202** (and from battery **102** to load **106**).

(30) Also, amplifier **272** can implement a feedback loop to set the gate voltage of transistor **202** to regulate the voltage $V_{sub.AC}$ across transistor **202** at a value equal to $V_{sub.F-reg}$ across different forward currents $I_{sub.F}$, to emulate a forward-biased diode as shown in FIG. 2A. Amplifier **272** can generate an output (e.g., a current, a voltage, etc.) that is linearly related to a difference between the anode-cathode voltage $V_{sub.AC}$ and a target forward voltage $V_{sub.F-reg}$ to adjust the on-resistance of the conduction channel of transistor **202**. The current provided by amplifier **272** can be converted to a voltage to set the gate voltage of transistor **202**, which in turn can set the on-resistance of transistor **202**. The on-resistance of transistor **202** can be adjusted, so the voltage $V_{sub.AC}$ across transistor **202** (which can be equal to a product between the on-resistance and the forward current) is maintained at the target forward voltage $V_{sub.F-reg}$. For example, if load **106** sinks more current, the voltage $V_{sub.AC}$ across transistor **202** can become larger than $V_{sub.F}$. In response, amplifier **272** can increase the gate-source voltage $V_{sub.GS}$ of transistor **202** (or the source-gate voltage $V_{sub.SG}$ if transistor **202** is PFET) to reduce the on-resistance of transistor **202**, to reduce the voltage $V_{sub.AC}$ back to $V_{sub.F-reg}$. However, if load **106** sinks less current, the voltage $V_{sub.AC}$ can decrease. In response, amplifier **272** can reduce the gate-source voltage $V_{sub.GS}$ of transistor **202** (or $V_{sub.SG}$ if transistor **202** is PFET) to increase its on-resistance, to increase the voltage $V_{sub.AC}$ back to $V_{sub.F-reg}$.

(31) With such arrangements, a voltage $V_{sub.AC}$ across transistor **202** can be maintained to emulate a forward-biased diode. The voltage provided by transistor **202** to load **106** can be maintained constant (or within a narrow range) and can be independent of forward current $I_{sub.F}$. This also allows the internal power supply (e.g., internal power supply **108**) of load **106** to provide

a stable supply voltage. Moreover, $V_{sub.AC}$ can be maintained at a low value to reduce power loss incurred by transistor **202**, especially when transistor **202** conducts a huge forward current $I_{sub.F}$ to load **106**.

(32) Referring again to FIG. 2D, RCB circuit **260** can receive low supply voltage $V_{sub.l}$ from local voltage generator circuit **256**, and forward conduction circuit **270** and inverter **282** can receive both low supply voltage $V_{sub.l}$ and high supply voltage $V_{sub.h}$ from local voltage generator circuit **256**. Specifically, RCB circuit **260** can operate within a voltage range below the anode voltage $V_{sub.A}$ to either disable transistor **202** by shorting the gate and source of transistor **202**, or releasing the gate of transistor **202**, therefore RCB circuit **260** can operate on low supply voltage $V_{sub.l}$ to reduce voltage stress and to improve reliability of the internal devices of RCB circuit **260**. Moreover, forward conduction control circuit **270** and inverter **282** can operate within a voltage range above the anode voltage $V_{sub.A}$. Such arrangements can increase gate overdrive voltage to enable transistor **202** while limiting the gate-drain voltage ($V_{sub.GD}$) and gate-source voltage ($V_{sub.GS}$) to reduce voltage stress across transistor **202**, which can improve the reliability of transistor **202**. Also, by operating forward conduction control circuit **270** and inverter **282** between the high supply voltage and the low supply voltage, the voltage swing in the devices of forward control circuit **270** and inverter **282** can be reduced, which can also reduce voltage stress and improve reliability of the internal devices of forward control circuit **270** and inverter **282**.

(33) Referring back to FIG. 1, while battery **102** outputs a DC voltage signal, reverse battery protection system **112** may receive AC voltage signals that are superimposed on the DC voltage output by battery **102**. The AC voltage signals can originate from various sources, such as electromagnetic interference, electrical noises, etc., from other electrical systems. For example, in a case where the system **100** is part of a vehicle, various electrical components, such as motors, switching power converters, etc., can generate periodic AC voltage signals that can be coupled into the wires that couple between battery **102** and electric power system **104**. The AC voltage signals can appear as part of the output voltage of battery **102** and can be superimposed with the DC voltage signal. The AC voltage signals can have a positive half-cycle and a negative half-cycle. In the positive half-cycle, the AC voltage signals add to the DC voltage signal to provide an increased anode voltage. In a negative half-cycle, in which the AC signals have negative voltages that subtract from the DC voltage signal to provide a reduced anode voltage. Body diode **210** of transistor **202** can switch between a forward-biased condition during the positive half-cycle and a reverse-biased condition periodically during the negative half-cycle. To protect load **106**, controller circuit can control the transistor to operate as a rectifier where the transistor is enabled under the forward-biased condition and is disabled under the reverse-biased condition.

(34) FIG. 3A illustrates an example of a DC voltage signal of battery **102** superimposed with AC voltage signals, and the rectification operation of reverse battery protection system **112**. Referring to the left side of FIG. 3A, graph **302** is an example graph of the anode voltage $V_{sub.A}$ with respect to time. The output voltage of battery **102** provides a DC input voltage (labelled “ $V_{sub.in_DC}$ ” in FIG. 3A) to reverse battery protection system **112**. As shown in graph **302**, the anode voltage $V_{sub.A}$ can include the DC component ($V_{sub.in_DC}$) as well as an AC component having an amplitude labelled “ $V_{sub.in_AC}$ ” superimposed with the DC component. The AC component can be attributed to the AC voltage signals. The AC component can be periodic and can have a positive half-cycle in which the AC component has a positive voltage that adds to the DC voltage and a negative half-cycle in which the AC component has a negative voltage that subtracts from the DC voltage. For example, the durations between times $T_{sub.0}$ and $T_{sub.1}$ and between times $T_{sub.2}$ and $T_{sub.3}$ can represent positive half-cycles, in which the AC component can add to the DC component, and output voltage of battery **102** (and anode voltage $V_{sub.A}$) exceeds the DC component $V_{sub.in_DC}$ and has a maximum voltage of $V_{sub.in_DC} + V_{sub.in_AC}$. Moreover, the duration between times $T_{sub.1}$ and $T_{sub.2}$ can represent a negative half-cycle, in which the AC components can be subtracted from the DC component, and output voltage of battery **102** (and

anode voltage $V_{\text{sub.A}}$) falls below the DC component $V_{\text{sub.in_DC}}$ and has a minimum voltage of $V_{\text{sub.in_DC}} - V_{\text{sub.in_AC}}$.

(35) Transistor **202** can switch between a forward-biased condition and a reverse-biased condition periodically between each half cycle of the AC signals superimposed with the DC voltage of battery **102**. During the positive half-cycles, the anode voltage (e.g., between $V_{\text{sub.in_DC}}$ and $V_{\text{sub.in_DC}} + V_{\text{sub.in_AC}}$) can be higher than the cathode voltage (e.g., $V_{\text{sub.in_DC}}$) and put transistor **202** in the forward-biased condition, and capacitor **226** can be charged up to $V_{\text{sub.in_DC}}$. During the negative half-cycles, the anode voltage (e.g., between $V_{\text{sub.in_DC}} - V_{\text{sub.in_AC}}$ and $V_{\text{sub.in_DC}}$) can be lower than the cathode voltage (e.g., $V_{\text{sub.in_DC}}$) and put transistor **202** in the reverse-biased condition. As part of the rectification operation to prevent a reverse current from flowing back from load **106** to battery **102** during the negative half-cycles, controller circuit **200** can enable transistor **202** during the positive half-cycles and disable transistor **202** during the negative half-cycles.

(36) Graph **304** of FIG. 3A shows an example graph of the cathode voltage $V_{\text{sub.C}}$ with respect to time as a result of a rectification operation performed by controller circuit **200** with transistor **202**. Specifically, during a positive half-cycle, such as between times $T_{\text{sub.0}}$ and $T_{\text{sub.1}}$, the anode-cathode voltage $V_{\text{sub.AC}}$ can exceed the forward conduction threshold voltage $V_{\text{sub.F-on}}$, and controller circuit **200** can enable transistor **202** to form a conduction channel between first current terminal **206** and second current terminal **208** to connect battery **102** to load **106**. Accordingly, the cathode voltage $V_{\text{sub.C}}$ across load **106** can track the anode voltage $V_{\text{sub.A}}$. The voltage $V_{\text{sub.AC}}$ can also be maintained at the target forward voltage $V_{\text{sub.F-reg}}$ by amplifier **272**. Accordingly, the cathode voltage $V_{\text{sub.C}}$ can have a DC component of $V_{\text{sub.out_DC}}$ which equals $V_{\text{sub.in_DC}} - V_{\text{sub.F-reg}}$, as well as an AC component having the same (or similar) amplitude $V_{\text{sub.in_AC}}$ as the AC signals at the anode. Within the positive half-cycle, the cathode voltage $V_{\text{sub.C}}$ can first increase from $V_{\text{sub.out_DC}}$, reaching a peak of $V_{\text{sub.out_DC}} + V_{\text{sub.in_AC}}$, and then decrease back to $V_{\text{sub.out_DC}}$.

(37) Moreover, during a negative half-cycle (e.g., between times $T_{\text{sub.1}}$ and $T_{\text{sub.2}}$), the anode voltage $V_{\text{sub.A}}$ is reduced by the AC component, while the cathode voltage $V_{\text{sub.C}}$ can be held at $V_{\text{sub.out_DC}}$ by holdup capacitor **226**, and the anode-cathode voltage $V_{\text{sub.AC}}$ can be below the reverse bias threshold voltage $V_{\text{sub.R}}$. This can cause controller circuit **200** to disable transistor **202** by removing/disabling the conduction channel between first current terminal **206** and second current terminal. Accordingly, the cathode voltage $V_{\text{sub.C}}$ can stop tracking the anode voltage $V_{\text{sub.A}}$ during the negative half-cycle. The negative half-cycle is followed by a subsequent positive half-cycle (e.g., between times $T_{\text{sub.2}}$ and $T_{\text{sub.3}}$), in which controller circuit **200** can enable transistor **202** again to allow a forward conduction from battery **102** to load **106**, and the cathode voltage $V_{\text{sub.C}}$ can track the anode voltage $V_{\text{sub.A}}$ again.

(38) The AC voltage signals that superimpose with the DC voltage signal output by battery **102** can have a high frequency, which can lead to high frequency changes in the input voltage to protection system **112**. FIG. 3B illustrates examples of AC voltage signals that may be received by protection system **112** in a vehicle application. For example, according to LV **124** and LV **148**, which define testing of electronic components for vehicles, the electric systems of a vehicle can have AC voltage signals of 2-6 peak-to-peak voltage (V_{pp}) at a frequency of 15 Hz to 200 kHz superimposed on the DC voltage signal output by a battery. Assuming the AC voltage signals have a frequency of 200 kHz and a cycle time of 5 micro-seconds (μs), to perform rectification, controller circuit **200** may need to be able to repeatedly enable transistor **202** within a positive half-cycle of 2.5 μs and disable transistor **202** within a negative half-cycle of 2.5 μs . Accordingly, it is desirable that the controller circuit **200** can have a short rectification response time, in order to enable and disable the transistor promptly in response to high-frequency AC voltage signals present at the battery output.

(39) But controller circuit **200** may have a long rectification response time, especially in enabling transistor **202** to start forward conduction, which makes it challenging to handle AC ripples up at a

frequency of 200 kHz and beyond. Specifically, as described above, to emulate a forward-biased diode having a constant forward voltage, controller circuit **200** (and forward conduction control circuit **270**) may include amplifier **272**, which can be linear amplifier such as an OTA, to implement a feedback loop to regulate the anode-cathode voltage $V_{sub.AC}$ across the transistor at the target forward voltage $V_{sub.F-reg}$. The output of the amplifier can be linearly related to $V_{sub.AC}$. But the amplifier may have a low gain (e.g., a low transconductance (g_m) for OTA) to improve loop stability, especially for low forward current/load current. The low gain can reduce the speed by which the amplifier can raise the gate voltage of the transistor, which in turn increases the time it takes for controller circuit **200** to enable transistor **202** (by forming a conduction channel between first current terminal **206** and second current terminal **208**) in the positive half-cycle of an AC ripple.

(40) FIG. 3C illustrates a graph **310** and a graph **320**. Graph **310** provides an example of variations of $V_{sub.AC}$ with respect to time, and graph **320** provides an example of variations of gate-source voltage $V_{sub.GS}$ of transistor **202** with respect to time, where transistor **202** is under the control of controller circuit **200**. Referring to graph **310**, $V_{sub.AC}$ can increase from a first voltage below $V_{sub.R}$ to a second voltage above $V_{sub.F-on}$ at time $T_{sub.0}$. In response to detecting that $V_{sub.AC}$ increases from below $V_{sub.R}$ to above $V_{sub.F-on}$, gate control circuit **250** to switch from disabling transistor **202** to enabling transistor **202** by changing the gate voltage.

(41) Referring to graph **320**, the gate-source voltage $V_{sub.GS}$ is initially at 0V and transistor **202** is disabled prior to time $T_{sub.0}$. In response to detecting the transition of $V_{sub.AC}$ at time $T_{sub.0}$, gate control circuit **250** may maintain $V_{sub.GS}$ at 0V, and transistor **202** can remain in the disabled state, until time $T_{sub.1}$. At time $T_{sub.1}$, gate control circuit **250** can start increasing $V_{sub.GS}$. At time $T_{sub.2}$, $V_{sub.GS}$ reaches the threshold voltage $V_{sub.th}$ of transistor **202**. In response to $V_{sub.GS}$ reaching $V_{sub.th}$, transistor **202** can form a conduction channel between first current terminal **206** and second terminal **208**, and transistor **202** is enabled. Accordingly, a total delay $T_{sub.D}$ has elapsed from the time $T_{sub.0}$ when the anode-cathode voltage $V_{sub.AC}$ transitions from lower than $V_{sub.R}$ to higher than $V_{sub.F-on}$ to the time $T_{sub.2}$ when transistor **202** is enabled.

(42) The delay $T_{sub.D}$ can be attributed to various sources. For example, a first part of the delay $T_{sub.D}$, between time $T_{sub.0}$ and $T_{sub.1}$, can be attributed to the delay incurred by RCB **260** in detecting the changes in $V_{sub.AC}$ and opening switch **266** (and closing switch **274**) to allow amplifier **272** to start increasing gate voltage $V_{sub.G}$. Moreover, a second part of the delay $T_{sub.D}$, between times $T_{sub.1}$ and $T_{sub.2}$, can be incurred by amplifier **272** in raising the gate voltage $V_{sub.G}$ as part of the feedback loop to regulate $V_{sub.AC}$ at $V_{sub.F-reg}$. The rate at which amplifier **272** increases $V_{sub.G}$ can be based on, for example, the gain (e.g., transconductance) of amplifier **272**, the capacitance of gate **204**, etc.

(43) The total delay $T_{sub.D}$ between time $T_{sub.0}$ and time $T_{sub.2}$ can represent a rectification response time of protection system **112**. Depending on various factors, such as comparator delay, the transconductance of amplifier **272**, the capacitance of gate **204**, etc., the rectification response time $T_{sub.D}$ can be near 2 μs . In a case where the AC ripples have a frequency of 200 kHz and beyond, the half-cycle period will be less than 2.5 μs . With a rectification response time that spans most of the positive half-cycle period, transistor **202** can remain disabled for most of the positive half-cycle period, which can prevent load **106** from receiving electric power from battery **102** during most of the positive half-cycle period. Accordingly, protection system **112** may be unable to perform the rectification operation in response to high-frequency AC signals present at the output of battery **102**.

(44) FIG. 4A through FIG. 4F illustrate examples of a controller circuit **400** that can address at least some of the issues. As shown in FIG. 4A, controller circuit **400** can be part of system **100** of FIG. 1 and can include RCB circuit **260** of FIGS. 2A through 2F. In addition, controller circuit **400** can include a forward conduction (FC) control circuit **402**, which can include an FC acceleration circuit

404, as well as amplifier 272 and switch 274 as described in FIGS. 2A through 2F. RCB circuit 260 and FC control circuit 402 can be part of a gate control circuit 410, and each of RCB circuit 260 and FC control circuit 402 is coupled with terminal 232 to set the gate voltage of transistor 202.

(45) FC control circuit 402 can set a gate voltage to form a conduction channel between first terminal 206 and second current terminal 208, to connect battery 102 to load 106. FC acceleration circuit 404 can first set the gate voltage to form the conduction channel, followed by amplifier 272 adjusting the gate voltage to regulate the $V_{sub.AC}$ voltage. Specifically, FC acceleration circuit 404 can include a switch 406 coupled between terminal 232 (and gate 204 of transistor 202) and a voltage reference 408 (labelled "VREF" in FIG. 4A). Switch 406 can be an NFET, a PFET, or a parallel combination of both. FC acceleration circuit 404 can receive an indication of a forward-biased condition. The indication can be based on control signal 280a. For example, based on detecting a transition of control signal 280a to a state to open switch 266, FC acceleration circuit 404 can determine that the forward-biased condition is present (with $V_{sub.AC}$ above $V_{sub.F-on}$), and close switch 406. The closing of switch 406 can connect voltage reference 408 with gate 204 of transistor 202 via terminal 232, which can raise the gate voltage (or reduce the gate voltage if transistor 202 is PFET) to enable transistor 202. Voltage reference 408 can provide a target voltage to enable transistor 202. After transistor 202 is enabled, if protection system 112 is still within the positive half-cycle, forward conduction circuit 402 can open switch 406, and provide a control signal 416 to close switch 274, which connects the output of amplifier 272 to terminal 232 and allows amplifier 272 to start a regulation loop to further adjust the gate voltage.

(46) Components of FC control circuit 402, including amplifier 472 and FC acceleration circuit 404, can operate on a high supply voltage $V_{sub.h}$ and a low supply voltage $V_{sub.l}$ generated by local voltage generator circuit 256 from anode voltage $V_{sub.A}$. For example, voltage reference 408 can be provided by the high supply voltage $V_{sub.h}$ if transistor 202 is an NFET. Voltage reference 408 can also be provided by low supply voltage $V_{sub.l}$ if transistor 202 is a PFET. The high supply voltage $V_{sub.h}$ and low supply voltage $v_{sub.l}$ can be configured based on, for example, a margin above (or below) a threshold voltage $V_{sub.th}$ of transistor 202 for forming the conduction channel to enable the transistor. In some examples, the high supply voltage $V_{sub.h}$ and low supply voltage $v_{sub.l}$ can also be configured based on a voltage stress threshold for the $V_{sub.GS}/V_{sub.SG}$ voltage, to improve reliability of transistor 202. Further, switch 406 can be controlled by a control signal that swings between $V_{sub.h}$ and $V_{sub.l}$ to reduce the on-resistance as well as the voltage stress on switch 406, which can improve both the bandwidth and reliability of switch 406.

(47) With the arrangements of FIG. 4A, forward conduction control circuit 402 need not rely on amplifier 272 to both start the enabling of transistor 202 and regulate the $V_{sub.AC}$ of transistor 202, and the reduced gain/transconductance of amplifier 272 (e.g., due to the constraint of loop stability) can have less impact on the delay in enabling transistor 202. Moreover, the speed of enabling transistor 202 can be further improved by increasing the bandwidth (e.g., by reducing the on-resistance) of switch 406. All these can reduce the rectification response time of controller circuit 400 and improve the handling of high-frequency AC signals at the output of battery 102.

(48) FIG. 4B illustrates a flowchart of an example method 420 performed by controller circuit 400 in controlling transistor 202. Method 400 can be performed after controller circuit 400 starts up and has not yet enabled transistor 202, and with battery 102 supplying a DC voltage to transistor 202. AC ripples may be coupled into and superimposed with the DC voltage, and appear as an AC component of the anode voltage $V_{sub.A}$ of transistor 202.

(49) In step 422, controller circuit 400 can determine an anode-cathode voltage ($V_{sub.AC}$) across transistor 202. Controller circuit 400 can monitor the anode voltage ($V_{sub.A}$) at terminal 230 and the cathode voltage ($V_{sub.C}$) at terminal 234. Controller circuit 400 can include a subtraction circuit (e.g., implemented using a differential amplifier) to subtract $V_{sub.C}$ from $V_{sub.A}$ to provide $V_{sub.AC}$.

(50) Controller circuit 200 can compare $V_{sub.AC}$ with forward conduction threshold voltage

V.sub.F-on using comparator **262a**, in step **424**. In addition, controller circuit **200** can also compare V.sub.AC with reverse conduction threshold voltage V.sub.R, in step **426**. V.sub.AC can exceed V.sub.F-on (and V.sub.R) during a positive half-cycle of the AC signals/ripples, and can be below V.sub.F-on or V.sub.R during a negative half-cycle of the AC signals/ripples, or when the battery is reverse connected.

(51) In step **428**, if V.sub.AC exceeds V.sub.F-on, RCB blocking logic **264** can output control signal **280a** to open switch **266** to disconnect RCB circuit **260** from the gate of transistor **202**. The opening of switch **266** can also enable FC control circuit **402** to set the gate voltage of transistor **202**. Moreover, based on control signal **280a** is in a state to open switch **266**, FC acceleration circuit **404** can close switch **406** to raise the gate voltage of transistor **202** (or reduce the gate voltage if transistor **202** is PFET). As to be described below, FC acceleration circuit **404** can close switch **406** for a pre-determined duration, or until the gate-source voltage exceeds the threshold voltage V.sub.th of transistor **202** (or falls below V.sub.th for PFET). Moreover, if V.sub.AC is below V.sub.F-on or V.sub.R (in steps **424** and **426**), RCB logic **264** can close switch **266** to bring the gate-source voltage V.sub.GS of transistor **202** to zero to disable the transistor, and switch **406** can be open, in step **430**.

(52) Referring back to step **428**, after transistor **202** is enabled by FC acceleration circuit **404** and a conduction channel is formed between first current terminal **206** and second current terminal **208**, controller circuit **400** can determine whether the positive half-cycle ends, in step **432**. The determination can be based on comparing a new V.sub.AC (obtained after transistor **202** is enabled) against V.sub.F-on and V.sub.R. If the new V.sub.AC is below either V.sub.F-on or V.sub.R, controller circuit **400** can determine that the positive half-cycle has ended, and can proceed to step **430** to disable transistor **202**.

(53) Moreover, if the new V.sub.AC is above V.sub.F-on in step **432**, which can indicate that the positive half-cycle has not yet ended, controller circuit **400** can start a regulation loop to regulate V.sub.AC at a target forward voltage V.sub.F-reg, in step **434**. As part of step **434**, FC acceleration circuit **404** can transmit control signal **416** to close switch **274**, which allows amplifier **272** to set the gate voltage of transistor **202** via terminal **232**. Amplifier **272** can also implement a feedback loop in which the output of amplifier **272** is linearly related to a difference between V.sub.AC and V.sub.F-reg, as described above. In step **444**, V.sub.AC reaches V.sub.F-reg, and can be regulated at V.sub.F-reg by amplifier **272**.

(54) FIGS. **4C** through **4F** illustrate examples of additional internal components of FC acceleration circuit **404**. Referring to FIG. **4C**, in addition to switch **406** and voltage reference **408**, FC acceleration circuit **404** can include a control circuit **450** and a driver circuit **452**. Control circuit **450** can receive control signal **280a**. In response to receiving control signal **280a**, control circuit **450** can provide control signal **416** to switch **274**, and a control signal **454** to switch **406**. Control circuit **450** can operate with the low supply voltage V.sub.l. Driver circuit **452** can receive control signal **454** and convert it to a control signal **456** that swings between a pre-determined minimum voltage v.sub.min and high supply voltage V.sub.h, and provide control signal **456** to switch **406**.

(55) FIG. **4D** and FIG. **4E** illustrate examples of internal components of control circuit **450**.

Control circuit **450** can include a start pulse circuit **460**, an end pulse circuit **462**, and a pulse generator **464**. Pulse generator **464** can generate control signal **454**, which can be in the form a pulse signal, to close switch **406** for a finite duration, with the start time and end time of the pulse signal being defined by, respectively, start pulse circuit **460** and end pulse circuit **462**. Start pulse circuit **460** can include an edge detection circuit **470** to detect a transition in control signal **280a** that indicates the start of a forward-biased condition. For example, the transition of control signal **280a** can be to a state to open switch **266**. Responsive to detecting the transition, start pulse circuit **460** can provide a start signal **472** that reflects the timing of the transition to pulse generator circuit **464**. In response to receiving start signal **472**, pulse generator circuit **464** can start the pulse of control signal **454**.

(56) In addition, end pulse circuit **462** can provide an end signal **474** to pulse generator circuit **464** to end the closing of switch **406**. End pulse circuit **462** can generate end signal **474** based on various techniques. Referring to FIG. **4D**, end pulse circuit **462** can include a delay circuit **476** to provide end signal **474** as a delayed version of start signal **472**, with the delay setting a width of the pulse of control signal **454**. Delay circuit **476** can include delay elements (e.g., buffers) to delay pull-up start signal **470**. In some examples, delay circuit **476** can include programmable delay elements such that the delay introduced to start signal **472** in generating end signal **474** is programmable. In such examples, delay circuit **476** can be coupled with a programming register to receive a setting for the delay.

(57) The delay of delay circuit **476** (and the pulse width) can be set based on various criteria. For example, the delay can be set based on the minimum half-cycle period of the AC ripples, such that the enabling of the pull-up path can start and end within a positive half-cycle period for the highest-frequency AC ripples to be rectified by controller circuit **400**. The delay can also be set based on other factors, such as the gate capacitance of transistor **202**, which can vary based on the dimension of the transistor. In some examples, the delay can also be set based on a target on-resistance of transistor **202** for the forward conduction operation. For example, based on the threshold voltage of transistor $V_{sub.th}$, a target gate voltage of transistor **202** can be determined, which can be a certain percentage (e.g., 90%) of the voltage provided by voltage reference **408**. Based on the gate capacitance of transistor **202** as well as the on-resistance of switch **406**, the time need to bring the gate voltage of transistor **202** to the target voltage can be determined, and the pulse width (and delay of delay circuit **476**) can be pre-configured accordingly.

(58) FIG. **4E** illustrates another example of pulse end circuit **462**. Specifically, pulse end circuit **462** can end the pulse and open switch **406** when the gate-source voltage $V_{sub.GS}$ goes above a threshold, such as threshold $V_{sub.th}$ of transistor **202** for forming a conduction channel between first current terminal **206** and second current terminal **208**. Referring to FIG. **4E**, pulse end circuit **462** can include a comparator **478** to compare the gate-source voltage $V_{sub.GS}$ (or $V_{sub.SG}$ for PFET) which can be obtained from a subtraction circuit that subtracts the source voltage $V_{sub.S}$ (obtained via terminal **230**) from the gate voltage $V_{sub.G}$ (obtained via terminal **232**). If $V_{sub.GS}$ rises above $V_{sub.th}$, comparator **478** can trip to generate pull-up end signal **472**, which can signal the end of the pull-up operation to pulse generator **466**.

(59) FIG. **4F** illustrates examples of internal components of driver circuit **450**, as well as switch **406** and voltage reference **408**, in a case where transistor **202** is an NFET. Referring to FIG. **4F**, switch **406** can be implemented using a PFET, coupled between voltage reference **408** and terminal **232**. Voltage reference **408** can be provided as the high supply voltage $V_{sub.h}$ from $V_{sub.high}$ terminal of local voltage generator circuit **256**. In addition, driver circuit **450** includes a level-shifter **480**, a load network **482**, and an NFET **484** coupled between load network **482** and terminal **230**. Level-shifter **480** can have a positive voltage supply terminal (labelled " $V_{sub.P}$ ") coupled with high supply voltage $V_{sub.h}$ (and voltage reference **408**), and a negative voltage supply terminal (labelled " V_n ") coupled with terminal **230**, which provides access to the anode voltage $V_{sub.A}$. Level-shifter **480** can receive, from pulse generator circuit **464**, control signal **454** that can swing between 0 to $V_{sub.l}$, and perform a level-shifting of control signal **454** to generate a control signal **486** with a voltage swing between $V_{sub.h}$ and $V_{sub.A}$, and drive NFET **484** with control signal **486**. Both control signals **452** and **486** can be pulse signals.

(60) In the example of FIG. **4F**, control signals **454** and **486** can be active-high pulse signals, with the high pulse defining the duration when switch **406** is closed. When NFET **484** is enabled by control signal **486**, NFET **484** can form a voltage divider with load network **482**, and the voltage divider can set the minimum voltage $V_{sub.min}$ of control signal **456** to PFET/switch **406**. Responsive to control signal **456** being at $V_{sub.min}$, PFET/switch **406** can be enabled/closed to connect gate **204** of transistor **202** to voltage reference **408**. When NFET **484** is disabled by control signal **486**, load network **482** can pull control signal **456** to the high supply voltage $V_{sub.h}$, and

PFET/switch **406** can be disabled/opened. Load network **482** can include a network of passive elements, such as resistors, Zener diodes, etc., to limit the voltage swing of the gate of pull-up transistor **480** to reduce voltage stress and to improve reliability.

(61) With the arrangements of FIG. **4F**, the gate voltage $V_{sub.G}$ can be set at a value higher than the anode voltage $V_{sub.A}$ (e.g., high supply voltage $V_{sub.h}$) to enable transistor **202**. As the anode voltage $V_{sub.A}$ is also the source voltage $V_{sub.S}$ of transistor **202**, the gate-source voltage $V_{sub.GS}$ can exceed the threshold voltage $V_{sub.th}$ to form the conduction channel between first current terminal **206** and second current terminal **208** of transistor **202**. Moreover, by using level-shifter **480** to level-shift a pulse signal that swings between low supply voltage $V_{sub.l}$ and ground to another pulse signal that swings between high supply voltage $V_{sub.h}$ and the anode voltage $V_{sub.A}$, the maximum voltages experienced by the transistor devices of pulse generator circuit **464**, as well as NFET **484** and PFET/switch **406**, can be reduced, which can reduce voltage stress and improve reliability of control circuit **400**. Further, as the anode voltage $V_{sub.A}$ (and source voltage $V_{sub.S}$) may vary based on the DC voltage output by battery **102**, generating the control signals and internal voltages based on the anode voltage $V_{sub.A}$ can ensure that the control signals and internal voltages have a deterministic relationship with the anode voltage $V_{sub.A}$ and source voltage $V_{sub.S}$, which can improve the predictability of control circuit **400**.

(62) FIG. **5** illustrates examples of a graph **510** of $V_{sub.AC}$ across transistor **202** with respect to time, and a graph **520** of the gate-source voltage $V_{sub.GS}$ of transistor **202** with respect to time, under the control of controller circuit **400**. Referring to graph **510**, $V_{sub.AC}$ increases from a first voltage below $V_{sub.R}$ to a second voltage above $V_{sub.F-on}$ at time $T_{sub.0}$. Referring to graph **520**, the gate-source voltage $V_{sub.GS}$ is initially at 0V and transistor **202** is disabled. It takes until time $T_{sub.1}$ when RCB circuit **260** generates control signal **280a** to disable switch **266**.

(63) Between times $T_{sub.1}$ and $T_{sub.2'}$, responsive to detecting that control signal **280a** is at a state to open switch **266**, FC acceleration circuit **404** can close switch **406** to raise (or reduce) the gate voltage of transistor **202**. The rate of change of the gate voltage can be impacted more by, for example, the on-resistance of switch **406**, the capacitance of gate **204**, etc., and less by the gain/transconductance of amplifier **472**. Accordingly, in FIG. **5**, the $V_{sub.GS}$ voltage can change at a higher rate and can reach the threshold $V_{sub.th}$ at time $T_{sub.2'}$, which is earlier than time $T_{sub.2}$ in FIG. **3C**. Accordingly, the total delay $T_{sub.D'}$ between times $T_{sub.1}$ and $T_{sub.2'}$ can also be reduced compared with $T_{sub.D}$ in FIG. **3C**. Switch **406** can be opened at time $T_{sub.2'}$. Between times $T_{sub.2'}$ and $T_{sub.3}$, amplifier **272** can operate a regulation loop to set the $V_{sub.GS}$ voltage in order to regulate the anode-cathode voltage $V_{sub.AC}$ at $V_{sub.F-REG}$.

(64) FIG. **6** is a flowchart of an example method **600** for performing a rectification operation by an electric power system coupled between a battery and a load, in a case where the DC voltage signal output by the battery is superimposed with AC signals. Method **600** can be performed by, for example, controller circuit **400** of FIGS. **4A** through **4F**. Although the method steps are described in conjunction with FIGS. **4A** through **4F**, any system configured to perform the method steps, in any suitable order, is within the scope of this description.

(65) At step **602**, controller circuit **400** can receive, via a first terminal (e.g., terminal **230**), a first voltage. The first terminal can be adapted to be coupled to a transistor's first current terminal (e.g., transistor **202**'s first current terminal **206**). The first current terminal can be an anode of a diode. The first current terminal can be a source if the transistor is NFET, and can be a drain if the transistor is PFET. The first current terminal can be adapted to be coupled to a battery (e.g., battery **102**). The first voltage can be an anode voltage ($V_{sub.A}$).

(66) At step **604**, controller circuit **400** can receive, via a second terminal (e.g., terminal **234**), a second voltage. The second terminal can be adapted to be coupled to the transistor's second current terminal. The second current terminal can be a cathode of the diode. The second current terminal can be a drain if the transistor is NFET, and can be a source if the transistor is PFET. The second current terminal can be adapted to be coupled to a load (e.g., load **106**). The second voltage can be

cathode voltage (V.sub.C).

(67) At step **606**, based on the first voltage and the second voltage, controller circuit **400** can close a switch (e.g., switch **406**) between a voltage reference (e.g., voltage reference **408**) and a third terminal (e.g., terminal **232**) of the controller circuit coupled to a gate of the transistor, in which a voltage of the gate is set by the voltage reference to form a conduction channel between the first current terminal and the second current terminal.

(68) Specifically, controller circuit **400** can include switch **406** coupled between terminal **232** (and gate **204** of transistor **202**) and voltage reference **408**. Voltage reference **408** can provide a target gate voltage to turn on/enable transistor **202**. The target gate voltage can be higher than the source voltage of transistor **202** if transistor **202** is NFET. The target gate voltage can also be lower than the source voltage of transistor **202** if transistor **202** is PFET. Voltage reference **408** can be provided by a local voltage generator circuit, such as local voltage generator circuit **256**, based on the anode voltage.

(69) If the anode-cathode voltage V.sub.AC exceeds reverse bias threshold voltage V.sub.R and forward conduction threshold voltage V.sub.F-on, RCB logic **264** can generate control signal **280a** to open switch **266**. Based on control signal **280a**, control circuit **400** can close switch **406** to connect the gate of transistor **202** to voltage reference **408** to raise (or reduce) the gate voltage. In some examples, controller circuit **400** can generate a pulse having a pre-determined pulse width to close switch **406**, with the start of the pulse triggered by control signal **280a**. In some examples, control circuit **400** can end the pulse (and open switch **406**) based on the gate-source voltage V.sub.GS (or source-gate voltage V.sub.SG for PFET) exceeding the threshold voltage V.sub.th of transistor **202** for forming the conduction channel between first current terminal **206** and second current terminal **208**.

(70) Step **606** can be performed as part of a rectification operation during the positive half-cycle of the AC signals. In some examples, if the positive half-cycle has not yet ended when the pulse ends and switch **406** is opened, controller circuit **400** can close switch **274** to enable amplifier **272** to further adjust the gate voltage to regulate V.sub.AC at the target forward voltage V.sub.F-reg. If the positive half-cycle has ended and the system is in the negative half-cycle of the AC signals, RCB logic **264** can close switch **266** to set the V.sub.GS voltage (or V.sub.SG voltage for PFET) to zero to disable transistor **202**.

(71) Accordingly, in some examples as described above, a protection system can include a controller circuit and a transistor, which can be a power transistor. Examples of protection system **100** are shown in FIG. 1 through FIG. 4B, whereas examples of controller circuit **400** are shown in FIGS. 4A through 4D. Referring to FIG. 4A, controller circuit **400** can include a first terminal (e.g., terminal **230**), a second terminal (e.g., terminal **234**), and a third terminal (e.g., terminal **232**). The first terminal is adapted to be coupled to a first current terminal of a transistor (e.g., first current terminal **206** of transistor **202**). The second terminal is adapted to be coupled to a second current terminal of the transistor (e.g., second current terminal **206** of transistor **202**). The third terminal is adapted to be coupled to a gate of the transistor (e.g., gate **204**). Controller circuit **400** can control the transistor to emulate an ideal diode, of which the first current terminal is the anode and the second current terminal is the cathode. The anode can be adapted to be coupled to a battery (e.g., battery **102**), and the cathode can be adapted to be coupled to a load (e.g., load **106**). In a case where the transistor is an NFET, the first current terminal can be a source and the second current terminal can be a drain. In a case where the transistor is a PFET, the first current terminal can be a drain and the second current terminal can be a source.

(72) Referring to FIG. 4A, controller circuit **400** can include gate control circuit **410**, which can include reverse current blocking (RCB) circuit **260** and forward conduction (FC) control circuit **402**. FC control circuit **402** can include a FC acceleration circuit **404** as well as amplifier **272**. Each of RCB circuit **260**, FC acceleration circuit **404**, and amplifier **272** can be coupled to terminal **232** (and gate **204** of transistor **202**) to set a gate voltage. RCB circuit **260** can set the gate-source

voltage $V_{sub.GS}$ (or source-gate voltage $V_{sub.SG}$ for PFET) to zero to disable transistor **202** if the anode-cathode voltage $V_{sub.AC}$ is below a reverse bias threshold voltage $V_{sub.R}$ or a forward conduction threshold voltage $V_{sub.F-on}$. RCB circuit **260** may include switch **266** coupled between terminals **230** and **232**, and may generate control signal **280a** to close switch **266** to set $V_{sub.GS}/V_{sub.SG}$ to zero to disable transistor **202**, or to open switch **266** to enable FC control circuit **402** to set the gate voltage. In some examples, RCB circuit **260** can disable transistor **202** in the negative half-cycles of the AC signals that superimpose with the DC signal output by the battery, as part of a rectification operation performed by controller circuit **400** with transistor **202**. (73) In addition, FC control circuit **402** can enable transistor **202** by raising (or reducing) the gate voltage to increase $V_{sub.GS}/V_{sub.SG}$ to enable transistor **202**, if $V_{sub.AC}$ is above $V_{sub.F-on}$. FC acceleration circuit **404** can include switch **406** coupled between voltage reference **408** and terminal **232**. If $V_{sub.AC}$ is above $V_{sub.F-on}$, FC acceleration circuit **404** can close switch **406** to connect terminal **232** (and gate **204**) to voltage reference **408** to change the gate voltage. The rate at which the gate voltage changes can be based on, for example, the on-resistance of switch **406**, the capacitance of gate **204**, etc., and can be independent from the gain/transconductance of amplifier **272**. FC acceleration circuit **404** can generate a pulse to close switch **406** for a finite duration. The start time of the pulse can be based on detection of $V_{sub.AC}$ being above $V_{sub.F-on}$ (e.g., based on control signal **280a**). The end time of the pulse can be based on a pre-determined delay from the start time, or based on the $V_{sub.GS}$ voltage (or $V_{sub.SG}$ voltage for PFET) exceeding a threshold voltage (e.g., $V_{sub.th}$ of transistor for forming the conduction channel). In some examples, FC control circuit **402** can enable transistor **202** in the positive half-cycles of the AC signals, as part of a rectification operation performed by controller circuit **400** with transistor **202**.

(74) Moreover, the output of amplifier **272** can be coupled to terminal **232** via switch **274**. After the pulse that closes switch **406** ends, if $V_{sub.AC}$ is still above $V_{sub.F-on}$ (e.g., the positive half-cycle has not yet ended), FC control circuit **402** can close switch **274** to allow amplifier **272** to further adjust the gate voltage by regulating $V_{sub.AC}$ at the target forward voltage $V_{sub.F-reg}$.

(75) In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action, then: (a) in a first example, device A is coupled to device B; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not substantially alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A. Also, in this description, a device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof. Furthermore, in this description, a circuit or device that includes certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package) and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, such as by an end-user and/or a third party.

(76) While certain components may be described herein as being of a particular process technology, these components may be exchanged for components of other process technologies. Circuits described herein are reconfigurable to include the replaced components to provide functionality at

least partially similar to functionality available before the component replacement. Components illustrated as resistors, unless otherwise stated, are generally representative of any one or more elements coupled in series and/or parallel to provide an amount of impedance represented by the illustrated resistor. For example, a resistor or capacitor illustrated and described herein as a single component may instead be multiple resistors or capacitors, respectively, coupled in series or in parallel between the same two nodes as the single resistor or capacitor. Also, uses of the phrase “ground voltage potential” in this description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. Unless otherwise stated, “about,” “approximately,” “nearly,” or “substantially” preceding a value means ± 10 percent of the stated value.

(77) Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. A system comprising: a transistor including a control terminal, a first terminal, a second terminal; and a control circuit including: a first circuit including: a first comparator including a first input, a second input, and an output; a second comparator including a first input, a second input, and an output; and a logic circuit coupled to the output of the first comparator and the output of the second comparator, the logic circuit including a first control output; and a second circuit including: a second control output coupled to the control terminal of the transistor; and a third circuit including an input, a first output, and a second output, wherein the input of the third circuit is coupled to the first control output, and the first output of the third circuit is coupled to the second control output.
2. The system of claim 1, wherein: the second circuit includes: a first switch including a first terminal coupled to the second control output; a third control output coupled to the first switch; and an amplifier including a first input, a second input, and an output coupled to a second terminal of the first switch.
3. The system of claim 2, wherein: a first reference is a target forward voltage.
4. The system of claim 2, wherein: the first circuit includes a second switch coupled to the first control output; a first terminal of the second switch is coupled to the second control output; a second terminal of the second switch is coupled to the first terminal of the transistor; and the third circuit includes a third switch coupled between a first voltage and the second control output.
5. The system of claim 4, wherein: the first comparator is coupled to a second voltage and a second reference; and in response to first comparator determining the second voltage is less than the second reference, the third circuit is configured to open the third switch.
6. The system of claim 4, wherein: the second comparator is coupled to a second voltage and a third reference; and in response to the second comparator determining the second voltage is greater than the third reference, the third circuit is configured to close the third switch.
7. The system of claim 4, wherein: the control circuit is configured to determine a positive half-cycle of a second voltage, the second voltage being coupled to the second input of the first comparator and the second input of the second comparator; and in response to determining the positive half-cycle of the second voltage is not ended, the third circuit is configured to output a signal on the third control output to close the first switch.
8. The system of claim 7, wherein: in response to determining the positive half-cycle of the second voltage has ended, the third circuit is configured to open the third switch.
9. The system of claim 1, wherein: a first voltage, coupled to the second input of the first comparator and coupled to the second input of the second comparator, is a difference voltage between a second voltage at the first terminal of the transistor and a third voltage at the second

terminal of the transistor; a first reference, coupled to the first input of the first comparator, is a reverse bias threshold voltage of a body diode of the transistor; and a second reference, coupled to the first input of the second comparator, is forward conduction threshold voltage.

10. The system of claim 1, wherein: the third circuit includes: a start pulse circuit including an input and an output, the input of the start pulse circuit coupled to the first control output; an end pulse circuit including an input and an output, the input of the end pulse circuit coupled to the output of the start pulse circuit; and a pulse generator circuit including a first input coupled to the output of the start pulse circuit, a second input coupled to the output of the end pulse circuit, and an output.

11. The system of claim 10, wherein: the start pulse circuit includes an edge detection circuit; and the end pulse circuit include a delay circuit.

12. A controller circuit comprising: a first circuit including: a first comparator including a first input, a second input, and an output; a second comparator including a first input, a second input, and an output; and a logic circuit coupled to the output of the first comparator and the output of the second comparator, the logic circuit including a first control output; and a second circuit including: a second control output; and a third circuit including an input, a first output, and a second output, wherein the input of the third circuit is coupled to the first control output, and the first output of the third circuit is coupled to the second control output.

13. The controller circuit of claim 12, wherein: the second circuit includes: a first switch including a first terminal coupled to the second control output; a third control output coupled to the first switch; and an amplifier including a first input, a second input, and an output coupled to a second terminal of the first switch.

14. The controller circuit of claim 13, wherein: the first circuit includes a second switch coupled to the first control output; a first terminal of the second switch is coupled to the second control output; and the third circuit includes a third switch coupled between a first voltage and the second control output.

15. The controller circuit of claim 14, wherein: the first comparator is coupled to a second voltage and a first reference; and in response to the first comparator determining the second voltage is less than the first reference, the third circuit is configured to open the third switch.

16. The controller circuit of claim 14, wherein: the second comparator is coupled to a second voltage and a second reference; and in response to the second comparator determining the first voltage is greater than the second reference, the third circuit is configured to close the third switch.

17. The controller circuit of claim 14, wherein: the control circuit is configured to determine a positive half-cycle of a second voltage, the second voltage is coupled to the second input of the first comparator and the second input of the second comparator; and in response to determining the positive half-cycle of the second voltage is not ended, the third circuit is configured to output a signal on the third control output to close the first switch.

18. The controller circuit of claim 17, wherein: in response to determining the positive half-cycle of the second voltage has ended, the third circuit is configured to open the third switch.

19. A method comprising: receiving, by a controller circuit, a first voltage; receiving, by the controller circuit, a first reference; in response to the first voltage being greater than the first reference, closing, by the controller circuit, a first switch between a second reference and a control output adapted to be coupled to a gate of a transistor; and in response to the first voltage being less than a third reference, opening, by the controller circuit, a second switch between an amplifier and the control output.

20. The method of claim 19, further comprising: determining, by the controller circuit, a positive half-cycle of the first voltage; in response to determining the positive half-cycle of the first voltage is not ended, initiating, by the controller circuit, a regulation loop configured to regulate the first voltage; and in response to determining the positive half-cycle of the first voltage has ended, opening, by the controller circuit, the second switch.

